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ROOT CAUSE ANALYSIS (RCA)

Company Name:	Direc Business Technologies Inc.
Purpose:	Root Cause Analysis (RCA)
Date:	June 18, 2026
Issue ID:	EIL_TEST_001
Issue:	Attendance records not updating after biometric synchronization
Assigned Consultant:	Jane Arianne A. Gaela
Approval Status:	<i>For Review</i>

ISSUE NO. 1

Attendance records not updating after biometric synchronization

PRODUCT People Navee HRIS

MODULE Attendance Management

CORE FUNCTION Biometric Synchronization Service

ROOT CAUSE ANALYSIS

1 ISSUE

The Human Resource department reported an issue where employee attendance records were not being updated after the scheduled biometric synchronization process. Employees were able to successfully clock in and clock out using the biometric devices. However, the attendance information was not properly displayed inside the HRIS Attendance Management module. The issue caused inconsistencies between the physical biometric logs and the records stored in the system database.

PRODUCT

People Navee HRIS

MODULEAttendance
Management**CORE FUNCTION**Biometric
Synchronization
Service**2 ROOT CAUSE****TECHNICAL ROOT CAUSE ANALYSIS**

After conducting an investigation, the development team identified that the failure occurred within the backend synchronization process responsible for transferring biometric device logs into the HRIS database.

The synchronization service encountered timeout errors whenever a large volume of employee attendance records was processed. The API connection was automatically terminated before the system completed the transaction.

Several contributing factors were discovered:

- The biometric API timeout configuration was too short for large attendance batches.
- The system processed records sequentially instead of using optimized batch processing.
- There was no automatic retry mechanism when synchronization failed.
- Error monitoring did not immediately notify administrators about failed synchronization attempts.

The issue was classified as a backend service reliability problem affecting data synchronization.

3 AFFECTED MODULE**AFFECTED SYSTEM COMPONENTS**

The issue affected multiple HRIS components that depend on accurate attendance information.

- Attendance Monitoring Dashboard
- Employee Daily Time Records

- Biometric Integration Service
- Payroll Attendance Validation
- Attendance Report Generation

Other HRIS modules remained operational and no employee profile information was affected.

4 IMPACT ANALYSIS

BUSINESS IMPACT ANALYSIS

The synchronization failure affected daily HR operations because attendance records required manual verification before processing.

Employees who successfully recorded their attendance through the biometric device appeared as absent or incomplete in the system.

Potential operational impacts included:

- Incorrect attendance monitoring results
- Possible payroll computation discrepancies
- Additional manual checking workload for HR personnel
- Delay in attendance approval workflow
- Reduced reliability of automated reporting

5 QUALITY GATE FINDINGS

QUALITY GATE — FIRST
PASS
Failed

SMOKE TEST — FIRST PASS
Passed

REOPEN COUNT
1

FC FAILED TESTING
Yes

QA STATUS
Validated After Fix

QA VALIDATED BY
**Jane Arianne A.
Gaela**

6 CORRECTIVE ACTION

CORRECTIVE ACTION IMPLEMENTED

The development team modified the synchronization service to improve performance, reliability, and failure handling.

Implemented solutions:

- Increased biometric API timeout limit from 30 seconds to 120 seconds.
- Implemented retry mechanism using exponential backoff strategy.
- Optimized attendance processing using database batch transactions.
- Added detailed error logging for failed synchronization attempts.
- Created validation checks before marking synchronization as complete.

After deployment, the synchronization process was tested with large attendance datasets and completed successfully.

7 PREVENTIVE ACTION

PREVENTIVE ACTION PLAN

Preventive improvements were planned to avoid similar issues in future releases.

- Implement automated synchronization monitoring alerts.
- Create daily attendance reconciliation reports.
- Add automated regression test cases for biometric integration.
- Perform scheduled performance testing on high-volume transactions.
- Improve API failure handling documentation.

8 OWNER REVIEW

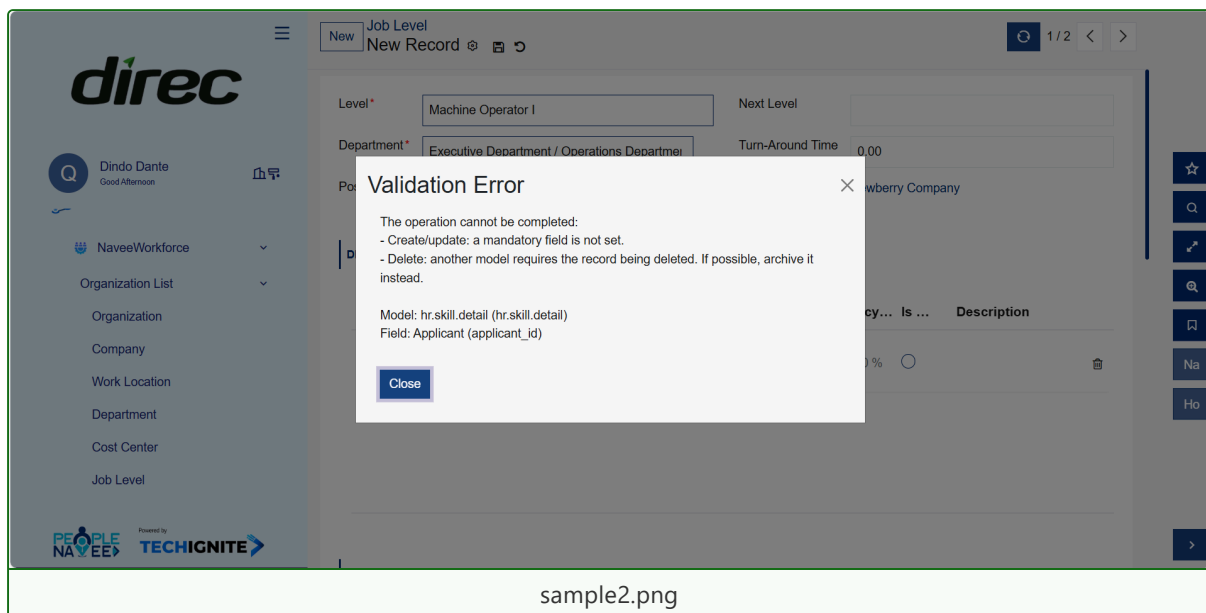
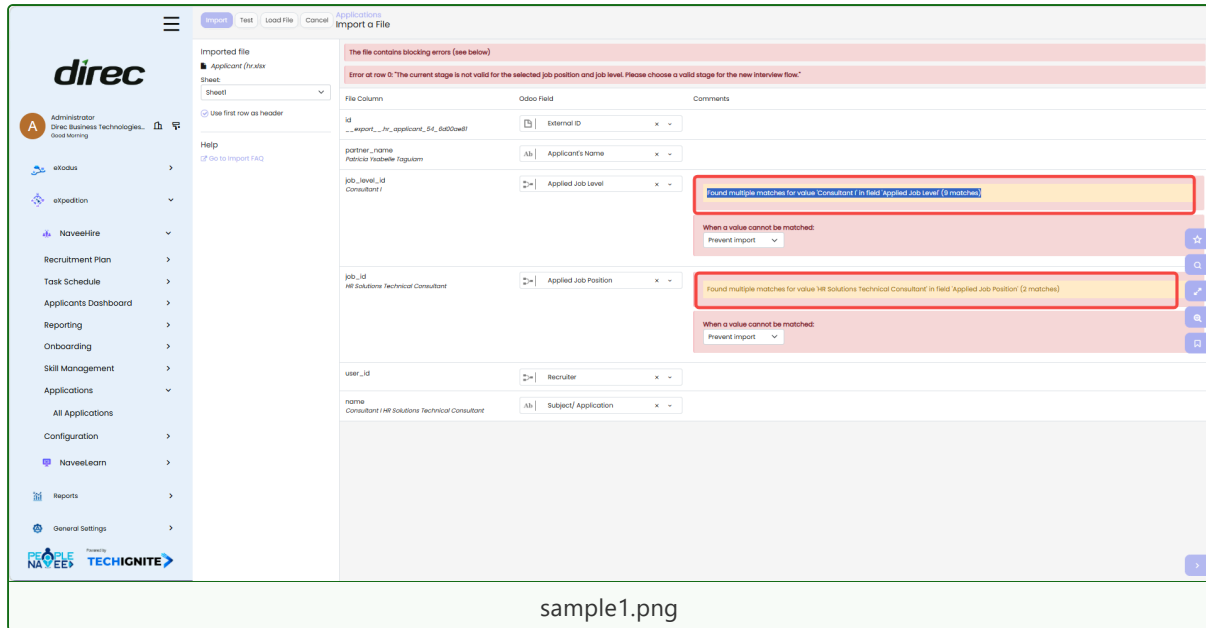
SYSTEM OWNER REVIEW

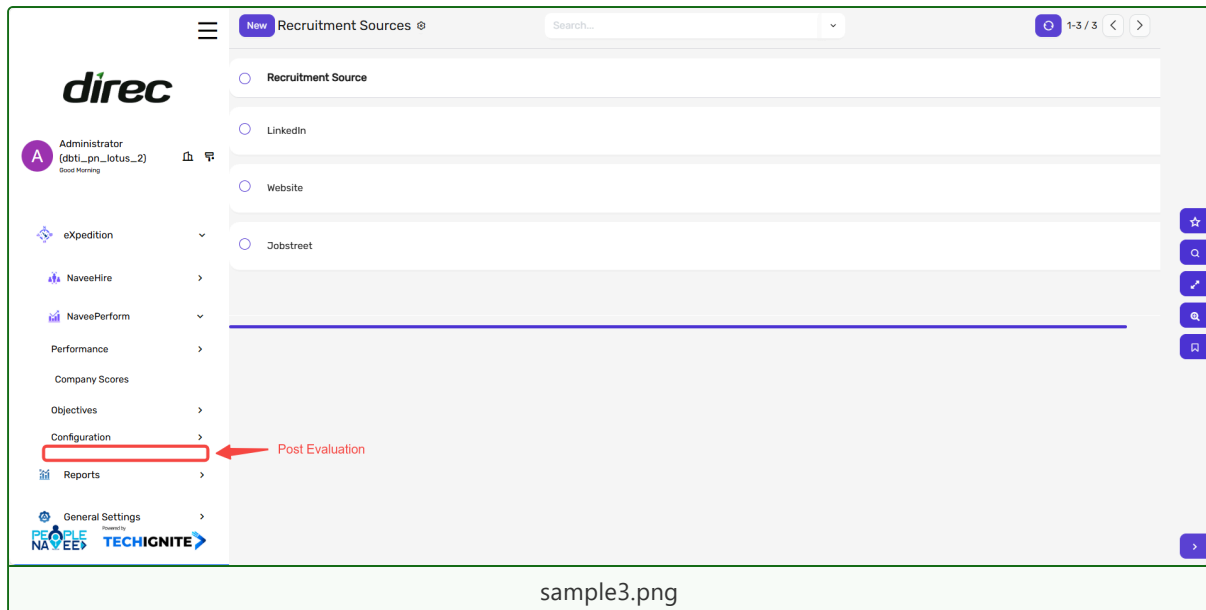
The system owner reviewed the implemented corrective actions, QA validation results, and testing evidence.

The solution successfully restored attendance synchronization functionality and improved system reliability.

The issue was approved for closure after confirming that no additional attendance inconsistencies were detected.

Attachments





APPROVAL SIGN-OFF

ROLE	NAME	SIGNATURE / DATE
Prepared By	Jane Arianne A. Gaela	
Reviewed By Developer	Jomar Talambayan	
Validated By QA	Joan Marie Peneda	
Approved By Owner	Emmanuel Manaloto	

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